

↳ Logical equivalences involving conditional statements :-

$$*1. \quad p \rightarrow q \equiv \neg p \vee q$$

$$*2. \quad p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$3. \quad p \vee q \equiv \neg p \rightarrow q$$

$$4. \quad p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$5. \quad \neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$6. \quad (p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$7. \quad (p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$8. \quad (p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$9. \quad (p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

① $p \rightarrow q \equiv \neg p \vee q$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

equivalent $\therefore$

⑥ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

$\neg p$	q	$\neg r$	$p \rightarrow q$	$p \rightarrow r$	$x \wedge y$	$q \wedge r$	$p \rightarrow z$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

$\hookrightarrow$ equivalent $\leftarrow$

↳ Logical equivalences involving biconditionals :-

*1. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

③. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

③	q	$\neg p$	$\neg q$	$\overset{\textcircled{x}}{p \wedge q}$	$\overset{\textcircled{y}}{\neg p \wedge \neg q}$	$x \vee y$	$p \leftrightarrow q$	$\neg(p \leftrightarrow q)$	$p \leftrightarrow \neg q$
T	T	F	F	T	F	T	T	F	F
T	F	F	T	F	F	F	F	T	T
F	T	T	F	F	F	F	F	T	T
F	F	T	T	F	T	T	T	F	F

equivalent.

equivalent.